Module Title	**Fundamentals of electricity and electronic**
Module Code	K72S015M01
Module Type	Basic Module (Technical)
Duration (Hrs.)	36 hour (T-18, P- 18)
Learning Outcomes	After completion of this module trainees will be able to • Explain the principles of electricity • Verify the functionality of the electronic components • Use testing tools like multimeter • Carry out Soldering/Desoldering work • Explain the importance of protecting the environment
Learning Content	Related Theory: • Principles of Electricity ○ Current, voltage, resistance ○ Ohm's law ○ Units & measurement ○ Electrical equations • Introduction to electronics • Electronic components ○ Resisters ○ Transistors ○ MOSFET ○ Thyristors ○ Capacitors ○ Diodes ○ Inductors ○ ICs • Introduction to fiber optic cables and optical transmission • Introduction to power supplies used in computer and related system • Heat sinking and thermal conducting methodology • Power protection devices (SPS, UPS, Surge arresters) • Noise filtering and shielding • Instruments used in electrical/electronic repairs ○ Soldering Iron / Hot gun ○ De soldering pump ○ Multimeter ○ Signal/Voltage generator ○ Oscilloscope ○ DC Power Supply • Behaviors of Electro Static Discharges (ESD) • Effects of Electro Magnetic Interferences (EMI) **(Duration-18 hour)**

	Practical: • Check AC, DC voltage and current, Resistance, Capacitance, Inductance using multimeter following safety procedures • Soldering/Desoldering selected components following safety procedures and environmental regulations **(Duration -18 hour)**
Teaching-Learning activities	• Illustrated talk • Demonstrations with Real objects • Video clips • Brainstorming session
Forms of assessment	• Formative Assessment + Summative Assessment

Module Title	**Computer fundamentals**
Module Code	K72S015M02
Module Type	Basic Module (Technical)
Duration (Hrs.)	18 hours (T12, P- 06)
Learning Outcomes	At the end of this module trainee should be able to; • Explain the basic functions of a computer system • Convert numbers between different number systems • Find required accurate information using relevant search engines • Send & receive Emails • Handle/use Social Network/ Media ethically
Learning Content	**Theory:** • Functions of the computer system ○ Input, Process, Output, Storage, Communication • Number systems ○ Binary, Octal, Decimal, Hexadecimal ○ Data representation methods (ASCII, Unicode) • Measuring storage capacities ○ bit, byte, kilobyte … etc. • Types of data ○ Analog, Digital data • Use of Internet & Email ○ Use of Web browsers ○ Use of general search engines ○ Use of AI search engines ○ Work with Email • Use of Social Networks/Media (E.g. Facebook, LinkedIn, Google Classroom, YouTube, Instagram ... etc.) ○ Work with communities ○ Ethical use of Social Networks/Media ○ Legal aspects of computer software ○ Software license, activation **(Duration-12 hour)**
	Practical • Create an Email account • Send email to few peoples using Cc/Bcc options • Create a five-page report on a given topic by the authorized person • Present the created report on an audience **(Duration-06 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Functions of Operating Systems, Utility software and Application Software**
Module Code	K72S015M03
Module Type	Basic Module (Technical)
Duration (Hrs.)	24 hour (T12, P- 12)
Learning Outcomes	At the end of this module trainee should be able to; • Create folders/Files • Prepare a formatted document • Prepare a worksheet • Make a presentation using presentation software
Learning Content	**Theory:** • Introduction to computer operating systems • Introduction to mobile operating systems (Android, iOS, etc.) • Classification of operating systems ○ Single user/multi user ○ Commercial/ Free and Open-Source Software (FOSS) ○ Command Line Interface (CLI)/ Graphical User Interface (GUI) • Versions, editions and distributions of operating systems • Basic functions of an operating system ○ Handling Folders (Create, Rename, Delete, Move) ○ Setting up configuration (Date & time, regional settings, etc.) ○ Find files and folders • Basic setting of internet connectivity (Connection and Password) • Introduction to word processing software ○ Page setup (paper size, orientation, margines) ○ Formatting text (Bold, underline, italic, etc.) ○ Tables ○ Images • Introduction to spreadsheet software ○ Page setup (paper size, orientation, margines) ○ Basic functions • Introduction to presentation software **(Duration-12 hour)**
	Practical • Create a given folder structure • Create a curriculum vitae (CV) using word processing software • Prepare a quotation using spreadsheet software • Make a presentation for a selected topic in an audience **(Duration-12 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Component of the computer system**
Module Code	K72S015M04
Module Type	Basic Module (Technical)
Duration (Hrs.)	24 hour (T12, P- 12)
Learning Outcomes	At the end of this module trainee should be able to; • Identify various components of a computer system • Explain the functions of each component • Check inter-compatibility of the selected component
Learning Content	**Theory:** • Introduction to different computer system o Personal computer/ Desktop o Work station and servers o Mobile computers/laptops • Components of a Computer system o Casing, power supply and power Adaptor o Motherboard o RAM/ROM o Processors (CPUs, Cooling system) o Adapter Cards and Expansion Slots o Hard Disk Drives and SSDs o Optical Storage Devices o Ports, Cables, and Adaptors • Peripheral devices o Input Devices o Output Devices (Monitors, Projectors, etc) o Storage devices o Communication devices **(Duration - 12 hrs.)**
	Practical • List the technical specification of the given component • Disassemble/ assemble components of a computer system following the relevant guidelines **(Duration-12hrs.)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Fundamentals of cyber security**
Module Code	K72S015M05
Module Type	Basic Module (Technical)
Duration (Hrs.)	18 hour (T-12, P- 06)
Learning Outcomes	At the end of this module trainee should be able to; • Scan the system for malware infections • Create a basic security policy document
Learning Content	**Theory:** • Introduction to cyber security • Malicious software (Malware) o Types of malwares o Degree of dangers from malware o Prevention methods of malware o Anti-malware • Introduction to network attacks • Software firewall • Introduction to social engineering attacks • Principles of security policy o User policy o Group policy **(Duration -12 hour)**
	Practical • Perform a system scan using a given anti-malware tool • Create a basic security policy document including user policy and group policy **(Duration-06hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Assemble Computers**
Module Code	K72S015M06
Module Type	Core Module
Duration (Hrs.)	30 hour (T-12, P- 18)
Learning Outcomes	At the end of this module trainee should be able to; • Identify assembling tools • Assemble a complete personal computer • Verify assemble components in BIOS settings
Learning Content	**Theory:** • Tools used in assembling process • Assembling order for various components • Safety guidelines should follow in assembling process • BIOS and CMOS settings and configuration ○ POST ○ UEFI/Legacy configurations ○ Boot order settings • Fault Replicable Unit (FRUs) • Remove and replacement procedures **(Duration: 12 hour)**
	Practical • Assemble a complete personal computer following standard steps • BIOS setting configuration and saving • Verify the functionality of assembled components using utility software **(Duration: 18 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Installing and Configuring Computer Software**
Module Code	K72S015M07
Module Type	Core Module
Duration (Hrs.)	24 hour (T-12, P- 12)
Learning Outcomes	At the end of this module trainee should be able to; • Install operating systems and essential software applications • Configure software settings to optimize functionality for specific user needs • Troubleshoot basic software installation and configuration issues
Learning Content	**Theory:** • Importance of software installation and configuration • Introduction to Software o Types of software (operating systems, application software, utility software) o Software licensing models (commercial, open-source) o System requirements and compatibility considerations o Understanding software architecture and components • Operating System Installation o Different operating systems (Windows, macOS, Linux) o Planning for OS installation (backing up data, system requirements) o Installation process for chosen operating system (physical media or digital download) o User permissions and administrative rights • Other Software Installation (Application, Utility, etc.) o Software installation methods (download, installation media) o Software licenses and end-user license agreements (EULAs) o Software management practices (uninstalling software, updating applications o Best practices for safe software downloads and avoiding malware • Software Updates o Importance of keeping software up-to-date o Configuring automatic updates o Manually updating the software • Software Licensing and Compliance o Types of software licenses (commercial, open-source, freeware) o Managing software licenses and subscriptions • System Maintenance Tools o Types of system utilities (cleanup tools, defragmentation, diagnostics) o Installing and configuring system maintenance tools o Regular system maintenance tasks

	• Software Configuration/Customization o Configuring software for specific user needs o Customizing software settings for specific user needs o Customizing program interfaces and preferences o User accounts and permissions within software • Installing and Configuring Security Software o Types of security software (antivirus, anti-malware, firewalls) o Installation and configuration of security applications o Running scans and configuring real-time protection **(Duration- 12 hour)**
	Practical • Install given operating systems on the assembled computer • Install given utility software • Install given application software **(Duration- 12 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Maintain computers and peripherals**
Module Code	K72S015M08
Module Type	Core Module
Duration (Hrs.)	24 hour (T-12, P- 12)
Learning Outcomes	At the end of this module trainee should be able to; • Perform preventive maintenance of the computer system • Perform upgrade/update computer system • Troubleshoot security and maintenance issues
Learning Content	**Theory:** • Importance of computer maintenance • Benefits of Preventive Maintenance • Tools and equipment for computer maintenance • Safe Handling Practices o Electrostatic Discharge (ESD) awareness and prevention techniques o Proper handling of computer components and peripherals o Safety precautions while working inside a computer case • Assembling and disassembling a computer o Step-by-step guide to assembling a computer o Disassembling and reassembling computer components o Safety precautions and best practices • Booting a computer o BIOS/UEFI Configuration o Update the Firmware • Software Tools and Utilities o Disk cleanup o Defragmentation o Other related utility software • Security Policy o Importance of computer security and data protection o Identifying and avoiding malware threats (viruses, spyware, etc) o Password management and best practices o Basic firewall functionalities • Security Threats o Types of Malwares o Anti-Malware Programs o Remediating Infected Systems • Firewalls o Software Firewalls o Installing and Configuring Firewalls • Web Security o Browser Extensions and Plugins

	○ Browser Privacy Settings ○ InPrivate Browsing ○ Pop-up Blocker • Protecting Data (Data Backup and Recovery) ○ Restore point ○ Importance of data backup ○ Methods and tools for data recovery ○ Data Encryption • Software Troubleshooting ○ Common software issues and solutions ○ Using system logs and error messages ○ Reinstalling and updating software • Hardware Troubleshooting ○ Identifying hardware failures ○ Diagnostic tools and techniques ○ Replacing faulty components • Network Troubleshooting ○ Common network issues and solutions ○ Tools for network diagnostics ○ Maintaining network security ○ User account • Troubleshooting and maintaining peripheral devices ○ Printers ○ Scanners ○ UPS ○ Other devices • Best Practices for Longevity ○ Power management and energy-saving tips ○ Environmental considerations ○ Documentation and record-keeping **(Duration: 12 hour)**

	Practical • Perform preventive maintenance for the given hardware components o Safely de-assemble o Clean the components • Perform upgrade/update computer system • Troubleshoot security and maintenance issues **(Duration: 12 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Troubleshooting and Repairing computer systems**
Module Code	K72S015M09
Module Type	Core Module
Duration (Hrs.)	42 hour (T-24, P- 18)
Learning Outcomes	At the end of this module trainee should be able to; • Identify Hardware & peripheral related problems • Identify Network & related devices problems • Identify OS and Application software issues and troubleshooting • Troubleshoot surveillance systems related issues
Learning Content	**Theory:** • Steps of troubleshooting ○ Troubleshooting flowcharts provided by the manufacturers • Common issues in computer components ○ Motherboard ○ Processor ○ RAM ○ VGA ○ Hard disk/SSD ○ Power Supply/Power Adaptor ○ Wireless Adaptor card ○ Sound Adaptor Card ○ Optical Drive ○ Keyboard ○ Touch pad ○ Interfaces and Ports • Common issues in peripheral devices ○ Monitors ○ Printers ○ Scanners ○ Projectors • Common issues in computer related additional devices ○ Biometric and access control devices ○ UPS/SPS ○ Routers/switches ○ Surveillance equipment • Common Operating System Related Issues ○ Startup/Boot Issues (BIOS) ○ Drivers ○ Fatal Error etc. ○ Upgrade/Update issues • Common application software related Issues ○ Compatibility issues ○ License issues

	• Importance of Ethical usage and Legal implications of unauthorized software/data usage **(Duration- 24 hour)**
	Practical • Perform troubleshooting following the instruction flowchart provided by the manufacturer • Perform troubleshooting for a given case **(Duration- 18 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Introduction to Data Communication and Computer Networks**
Module Code	K72S015M10
Module Type	Core Module
Duration (Hrs.)	24 hour (T-12, P- 12)
Learning Outcomes	After completion of this module the trainee will be able to: • Understand networking basics, topologies, and data transmission media. • Identify hardware components (switches, routers, access points, patch panels).
Learning Content	Theory: • Introduction to computer network • Data communication fundamentals o Data Transmission Methods ▪ Analog, Digital o Communication Types ▪ Simplex, Half Duplex, Full Duplex o Communication Media ▪ Guided (wired) - Co-Axial - Twisted wire pair (STP, UTP, Cat5e, Cat6, etc.) - Fiber optics ▪ Unguided (Wireless) - Radio waves - Microwaves - Satellite - Infrared - Wi-Fi, Bluetooth o Mobile communication technology ▪ 1G, 2G, 3G, 4G. 5G • Network Topologies o Bus, Ring, Star, Mesh, Tree, Hybrid • Types of Networks o LAN, WAN, WLAN • Network devices o Switches o Routers o Access point **(Duration- 12 hour)**
	Practical: Identify Network devices and components for a given scenario **(Duration- 12 hour)**

Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Network Device Installation and Configuration**
Module Code	K72S015M11
Module Type	Core Module
Duration (Hrs.)	72 hour (T-24, P- 48)
Learning Outcomes	After completion of this module the trainee will be able to: 1. Backup of device configurations 2. Firmware updates and patch management 3. Preventive maintenance scheduling 4. Monitoring device health and performance 5. Troubleshooting and resolving post-installation issues
Learning Content	Theory: • **Introduction to Network Installation** o Overview of network installation processes o Types of network solutions ▪ LAN, MAN, WAN ▪ VLAN, WLAN ▪ Enterprise networks o Network architecture models: client–server, peer-to-peer, hybrid • **Network Installation** o Site survey and inspection: ▪ Accessibility ▪ Power availability ▪ Space for racks and devices ▪ Environmental conditions (temperature, airflow, dust) ▪ Safety compliance o Pre-installation checks: grounding, continuity, wiring conditions o Completing prerequisites: casings, trunking, rack mounting, floor/ceiling preparation • **Structured Cabling and Physical Installation** o Principles of structured cabling o Horizontal and vertical cabling concepts o Cable installation best practices: ▪ Avoiding tension, sharp bends, electrical interference ▪ Maintaining separation from power lines ▪ Following color codes (T568A/T568B) o Terminating cables onto patch panels, faceplates, and connectors o Installing and mounting network devices (switches, routers, servers, access point) o Rack organization: ▪ Cable management ▪ Ventilation

	▪ Power distribution units (PDUs) o Labeling cables, ports, racks, and devices correctly o Creating initial physical documentation and diagrams • **Network Device Configuration** o **IP Addressing** ▪ Static vs dynamic addressing ▪ IPv4 and IPv6 addressing basics ▪ Subnetting and VLAN-specific addressing ▪ Assigning IP addresses to end devices and network equipment o **Switch Configuration** ▪ Access ports vs trunk ports ▪ VLAN creation and tagging ▪ Port security concepts o **Router Configuration** ▪ Basic routing configuration ▪ Static routing setup ▪ Introduction to dynamic routing protocols (RIP, OSPF, EIGRP) ▪ NAT configuration o **Wireless Configuration** ▪ SSID creation and broadcast settings ▪ Wireless security protocols (WPA2/WPA3) ▪ Channel selection and interference reduction ▪ Configuring centralized or standalone access points o **Network Services** ▪ DHCP configuration (scopes, reservations) ▪ DNS configuration (records, resolution basics) ▪ VPN basics (site-to-site, remote access) ▪ Firewall rule configuration and access control lists (ACLs) o **Configuration Documentation** ▪ Recording device configurations ▪ Updating IP address tables ▪ Version control for configuration files ▪ Network topology map updates • **Network Testing & Validation** o Connectivity testing tools and commands: ▪ Ping (host reachability) ▪ Traceroute (path tracing) ▪ Nslookup (DNS verification) ▪ Netstat (port and connection checks)

	○ Testing physical media using cable testers and certification tools ○ Validating VLANs, routing tables, and DHCP/DNS operations ○ Wireless coverage and signal strength testing ○ Troubleshooting common installation issues: ▪ IP conflicts ▪ Faulty terminations ▪ Incorrect VLAN assignments ▪ Routing misconfigurations • **Finalizing the Network Installation** ○ Preparing installation and configuration reports ○ Handover procedures to client or management ○ Backup of configuration files ○ Ensuring security hardening and user access control ○ Post-installation support and maintenance plan • **Tools & Safety in Network Installation** ○ Tools for installation: ▪ Crimping tools, punch-down tools, cable testers, tone generators ▪ Labeling machines, drillers, ladders ○ Safety guidelines: ▪ Electrical safety ▪ Ladder and height safety ▪ Dust and ventilation control ▪ PPE usage ○ Environmental considerations in network rooms (cooling, humidity, airflow) **(Duration- 24 hour)**
	Practical: 1. Prepare a network installation plan following a site survey. 2. Mount and cable a switch, router, and access point in a rack. 3. Setting-up a DHCP server 4. Document the network connectivity test results. **(Duration- 48 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Fundamentals of Computer Networking**
Module Code	K72S015M12
Module Type	Core Module
Duration (Hrs.)	42 hour (T-24, P- 18)
Learning Outcomes	After completion of this module the trainee will be able to: • Understand networking basics, topologies, and transmission media. • Identify hardware components (switches, routers, access points, patch panels). • Perform structured cabling and basic LAN setup.
Learning Content	Theory: • Reference model o OSI model o TCP/IP suite • IP addressing o IPv4 o IPv6 • Subnetting o Classful, Classless • Structured cabling standards • VLAN basics • Cable testing and documentation • Installation safety and ESD precautions **(Duration- 24 hour)**
	Practical: • Create LAN connections. • Terminate cables (RJ45, patch panel). • Configure IP addresses. • Test connectivity using ping/traceroute. **(Duration- 18 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Network System Setup and Configuration**
Module Code	K72S015M13
Module Type	Core Module
Duration (Hrs.)	60 hour (T-24, P- 36)
Learning Outcomes	At the end of this module trainee should be able to; • Perform network cabling and configure network devices • Install and configure Network Operating Systems (NOS) • Configure network settings for IP addressing, subnet masks, and network protocols • Handle network resources like users, groups, and permissions. • Troubleshoot common network connectivity and performance issues • Apply security best practices for network protection
Learning Content	**Theory:** • Introduction to Networking o Importance of network systems o Network fundamentals (protocols, devices, media) o Network topologies (bus, star, mesh, etc.) o OSI and TCP/IP Model and its layers o Networking terminology and acronyms (IP address, subnet mask, gateway, etc.) • Network Cabling and Devices o Types of network cables o Cable crimping and termination techniques o Network devices (routers, switches, access points) o Functions of different network devices • Wireless Networking o Understanding wireless networking standards (Wi-Fi, Bluetooth) o Setting up wireless access points (WAPs) o Configuring wireless security (WPA2, WPA3) • IP Addressing and Subnetting o Understanding IP addressing (IPv4, Ipv6) o Subnetting and subnet masks o Classful and classless addressing schemes o Private and public IP addresses o Calculating and assigning IP addresses

	• Installing Network Hardware o Installing and configuring network interface cards (NICs) o Setting up routers and switches o Connecting network cables and testing connections • Network Operating Systems o Installing and configuring network operating systems o Setting up network services (DHCP, DNS, FTP) o Managing users and permissions • Sharing of devices o File sharing o Printer sharing • Network Security Fundamentals o Understanding network security principles o Implementing firewalls and intrusion detection systems (IDS) o Configuring VPNs and secure remote access • Network Monitoring and Management o Using network monitoring tools o Managing network performance and bandwidth o Troubleshooting network connectivity issues • Network Troubleshooting Techniques o Identifying and diagnosing network problems o Using diagnostic tools and commands (ping, traceroute, nslookup) o Resolving common network issues • Cloud services o Cloud storage services o Setup and configuration of cloud storage **(Duration- 24 hour)**
	Practical • Lay network cable, crimping keystone, connect to patch panel • Test network cable for connectivity • Setup network configuration for a given network • Create, configure and map cloud storage service **(Duration- 36 hour)**
Teaching-Learning activities	• Assignments • illustrated talk • Practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Maintenance of Computer Networks**
Module Code	K72S015M14
Module Type	Core Module
Duration (Hrs.)	60 hour (T-24, P- 36)
Learning Outcomes	After completion of this module the trainee will be able to: 1. Identify and perform corrective maintenance based on user requirements and warranty checks. 2. Plan and execute preventive maintenance activities according to schedules and SOPs. 3. Establish, execute, verify, and restore backup operations. 4. Document all maintenance activities and manage e-waste responsibly. 5. Use diagnostic and monitoring tools to verify network performance and connectivity.
Learning Content	Theory: • **Introduction to Network Maintenance** o What is Network maintenance? o Common causes of network issues ▪ High bandwidth usage ▪ Faulty hardware ▪ High CPU utilization ▪ Poor connectivity o Network maintenance plan o Network reliability and security o Roles and responsibilities of a network maintenance technician o Types of Network maintenance ▪ Corrective, Preventive and Predictive maintenance • **Corrective Maintenance** o Corrective maintenance request or complaint handling procedures o Warranties and vendor guidelines o Tools, equipment, software and spare parts o Corrective maintenance procedures o Corrective maintenance documentation and log maintenance • **Preventive Maintenance** o Maintenance schedules ▪ Prepare maintenance schedule ▪ Review maintenance schedule with users o Inspecting, cleaning and testing physical components ▪ Cables, switches, routers, access points, etc. o Software and firmware updates o Post-maintenance performance test o Preventive maintenance documentation • **Backup and Recovery Management**

	○ Importance of backups ○ Types of backups ▪ Incremental, Full, Differential, Configuration only ▪ Backup media (Online, Offline, Realtime, Cloud) ▪ Automated, Manual backup ○ Backups and restoration procedures ○ Backup and restoration documentation and log maintenance • **Tools and Best Practices** ○ Network diagnostic tools (ping, traceroute, network scanners, etc.) ○ Monitoring tools (SNMP, Wireshark, etc.) ○ Best practices in cable management, device labeling, and workspace safety ○ E-waste management and disposal procedures • **Security Maintenance** ○ Awareness of security policies and firewalls ▪ Firewalls rules and configuration methods ▪ Password control policies, access control lists (ACLs) maintenance ○ Logs maintenance controlling for: ▪ Unauthorized access ▪ Malware activities ▪ Brute-force or scanning attempts ○ Antivirus and Event Data Record (EDR) maintenance **(Duration- 24 hour)**
	Practical: 1. Hands-on practice to remove, clean, refix routers, switches, access points and other relevant devices of computer network. (Should maintain the nodes numbering as it is. 2. Perform a full network health check using monitoring tools. 3. Create a configuration backup and restore 4. Prepare a maintenance report with maintenance logs and warranty details **(Duration- 36 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Network Security Setup and Configuration**
Module Code	K72S015M15
Module Type	Core Module
Duration (Hrs.)	60 hour (T-24, P- 36)
Learning Outcomes	After completion of this module the trainee will be able to: 1. Identify and mitigate network security threats 2. Configure secure routers, switches, firewalls, and wireless networks 3. Implement ACLs and VLAN-based segmentation 4. Monitor and respond to security events 5. Maintain accurate documentation of network security configurations
Learning Content	Theory: ○ **Introduction to Network Security** ○ **Definition and purpose of network security** ○ **Common threats and vulnerabilities**: ▪ Malware, ransomware, phishing ▪ Unauthorized access ▪ DoS/DDoS attacks ▪ Insider threats ○ **CIA Triad:** ▪ Confidentiality ▪ Integrity ▪ Availability ○ **Security policies and governance standards:** ▪ ISO 27001 ▪ PCI-DSS basics ▪ Organizational security procedures ○ **Security Components and Devices** ○ **Firewalls** ▪ Hardware vs software firewalls ▪ Stateful inspection ▪ Stateless filtering ○ **Routers** ▪ Basic security functions ▪ Routing security (ACLs, filtering) ○ **Switches** ▪ Port security ▪ MAC address table verification ▪ MAC address filtering ▪ VLAN segmentation ○ **Access Points**

	▪ Wireless security mechanisms ▪ Authentication modes (Open, WPA2/WPA3, Enterprise) o **Network Access Control** o **Authentication, Authorization, Accounting (AAA)** o **User and device authentication** ▪ Password policies ▪ Multi-factor authentication (MFA) o **802.1X Network Access Control** ▪ RADIUS server concept ▪ Supplicant, authenticator, authentication server o **Role-based access control (RBAC)** ▪ Guest network access control and captive portals o **Firewall Configuration and Implementation** ▪ Types of firewall rules: – Allow, deny, reject – Inbound vs outbound rules ▪ Creating firewall rule sets: – Source/destination IP – Port numbers – Protocol (TCP/UDP/ICMP) ▪ Best practices for firewall configuration: – Default deny policy – Least privilege rule creation – Logging and monitoring ▪ NAT and PAT ▪ DMZ (Demilitarized Zone) o **Secure Router & Switch Configuration** ▪ Router hardening: – Disable unused services – Secure management access (SSH vs Telnet) – Password encryption – Banner warnings ▪ Switch security enhancements: – Port security: – Sticky MAC – Limiting number of MACs – Violation actions ▪ VLAN segmentation to isolate traffic ▪ Dynamic ARP Inspection (DAI) ▪ BPDU Guard and Loop Prevention ▪ Updating firmware and OS patches

	○ **Wireless Network Security** ▪ Wireless encryption standards: – WEP (obsolete) – WPA, WPA2, WPA3 ▪ SSID configuration: – Broadcast control – Hiding vs not hiding SSID ▪ Securing wireless channels: – Avoiding interference – Selecting optimal channels ▪ Enterprise Wi-Fi Security: – WPA2/WPA3-Enterprise with RADIUS – Certificate-based authentication ▪ Guest Wi-Fi isolation techniques ○ **Endpoint & Perimeter Security** ▪ Antivirus and EDR tools ▪ Patch management ▪ Secure remote access: – VPN configuration basics – SSL VPN vs IPsec VPN ▪ Email and web filtering ▪ Intrusion Detection and Prevention Systems (IDS/IPS): ▪ Signature-based detection ▪ Anomaly-based detection ▪ Inline vs passive modes ○ **Network Monitoring & Logging** ▪ Importance of logging: – Security auditing – Forensics ▪ Syslog configuration and log levels ▪ Network monitoring tools: – SNMP – Wireshark – Traffic analyzers ▪ Detecting unusual activity: – Traffic spikes – Unauthorized access attempts – Port scanning

	○ **Security Hardening Procedures** ▪ Hardening network devices: – Disable unused ports/services – Use strong encryption (SSH, TLS) – Set strong admin passwords ▪ Hardening servers and endpoints ▪ Physical security practices: – Locking racks – Controlled access areas – CCTV and environmental monitoring **(Duration- 24 hour)**
	Practical: • Configure firewall rules and implement Access Control Lists (ACLs) to secure network traffic ○ Configure inbound and outbound firewall rules: ○ Implement ACLs on a router/switch to: ○ Test the configuration ○ Document all rules and ACL configurations. • Configure and harden routers, switches, and wireless access points for network security. ○ **Router Hardening:** ▪ Disable unused services (Telnet, HTTP). ▪ Enable SSH and configure strong encrypted passwords. ▪ Set up login banners. ○ **Switch Security:** ▪ Configure port security (limit MAC addresses, sticky MAC). ▪ Create VLANs for network segmentation. ▪ Enable DHCP snooping and Dynamic ARP Inspection. ○ **Wireless Security:** ▪ Configure WPA3 Wi-Fi with strong password. ▪ Assign VLAN for guest network isolation. ▪ Adjust wireless channels to minimize interference. ▪ Test network connectivity, VLAN isolation, and wireless security. • Monitor network activity and detect security events using logging and monitoring tools ○ Configure syslog on routers, switches, and firewalls to send logs to a central server. ○ Monitor and analyze logs for: ▪ Unauthorized access attempts

	▪ Firewall rule violations ▪ Port scans or suspicious traffic ○ Use a network monitoring tool (e.g., Wireshark, PRTG) to: ▪ Capture and analyze traffic for anomalies ▪ Identify potential security threats ○ Create a security report summarizing findings and corrective actions. **(Duration- 36 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Installation and Maintenance of Surveillance and Biometric Security Systems**
Module Code	K72S015M16
Module Type	Core Module
Duration (Hrs.)	60 hour (T-24, P- 36)
Learning Outcomes	After completion of this module the trainee will be able to: 1. Install surveillance and biometric equipment safely 2. Configure cameras, recorders, and biometric terminals 3. Integrate CCTV with access control systems 4. Perform system testing and troubleshooting 5. Maintain and update security systems 6. Prepare documentation and reports
Learning Content	Theory: **Biometric System Components & Architecture** • Types of biometric systems: o Fingerprint scanners o Facial recognition terminals o Iris scanners o RFID card/badge access systems • Biometric database basics • Authentication modes: o 1:1 Verification o 1: N Identification • Controllers and panels (access control units) • Software platforms for biometric attendance and access management • Integration with door locks, turnstiles, gates **Introduction to Surveillance Systems** • Purpose and applications of surveillance systems • Industry uses: o Security monitoring o Evidence recording • Overview of technologies: o CCTV (Analog, HD, IP) o Sensors • Benefits, limitations, and ethical considerations • Legal guidelines and privacy regulations related to surveillance **Surveillance System Components & Architecture** • Camera Types: o Dome, Bullet, PTZ (Pan–Tilt–Zoom) o Day/night cameras, IR cameras o IP vs Analog cameras • Recording Systems: o DVR (Digital Video Recorder) o NVR (Network Video Recorder) o Hybrid recorders • Storage Technologies**:** o HDD types (Surveillance-grade) o RAID configurations

	○ Cloud backup basics • Network Components: ○ PoE switches ○ Cables (Cat6, coaxial) ○ Routers, Access Points • Software Components: ○ VMS (Video Management Software) ○ Mobile/remote monitoring apps • Surveillance system requirement identification ○ Client goals and security needs identification ○ Coverage areas: ▪ Entrances/exits ▪ Perimeter ▪ High-value zones ▪ Blind spot detection ○ Camera specifications ▪ Camera quantity ▪ Lens type (wide-angle, varifocal) ▪ Storage technology ▪ Power sources ○ Environmental factors: ▪ Lighting conditions ▪ Indoor/outdoor placement ▪ Weatherproof and vandal-proof requirements ○ Cable installation and jointing ○ Biometric installation considerations: ▪ User traffic calculation ▪ Height and mounting location • Installation of Surveillance Systems ○ Mounting cameras: ▪ Proper height and angles ▪ Avoiding backlight, glare ▪ Securing with appropriate brackets ○ PoE setup for IP cameras ○ DVR/NVR installation: ▪ Rack mounting ▪ HDD installation & formatting ▪ Cooling & ventilation ○ Power distribution: ▪ Centralized power supply vs individual adapters ▪ Surge protection ▪ UPS integration • Configuration of Surveillance Systems ○ IP addressing for cameras & NVR ○ Camera configuration: ▪ Resolution settings ▪ Frame rates ▪ Motion detection zones ▪ Image adjustments (brightness, WDR, IR settings)

	○ Network configuration: ▪ Port forwarding for remote access ▪ DDNS configuration ▪ VLAN segregation for security ○ VMS setup: ▪ Adding cameras ▪ Creating camera groups ▪ Recording schedules ▪ User permissions and roles ○ Mobile app setup for remote viewing • Installation of Biometric Systems ○ Device mounting and power connection ○ Door lock integration (magnetic lock, strike lock) ○ Cabling for access control: ▪ Exit button wiring ▪ Door sensor setup ○ Network setup for biometric terminals: ▪ Static IP assignment ▪ Server communication ▪ Data synchronization ○ Defining user roles and access levels ▪ Fingerprint, face, card registration • System Integration ○ Integrating CCTV with access control ○ Alerts and triggers: ▪ Door forced alerts ▪ Camera activation on motion/event ○ Third-party integration possibilities: ▪ Alarm systems ▪ Fire detection panels ○ API and cloud integrations • Troubleshooting common issues: ○ No video ○ Camera offline ○ Poor image quality ○ Biometric mismatch ○ Door lock not responding • Maintenance of Surveillance & Biometric Systems ○ Periodic cleaning of cameras and lenses ○ Storage health check (HDD/SSD) ○ Firmware updates for cameras & biometric devices ○ Log review: ▪ Security events ▪ Access attempts ○ Database backup and recovery procedures ○ Checking power systems (UPS, power supplies) ○ Performance optimization: ▪ Adjusting video settings

	▪ Clearing old logs ▪ Re-calibrating biometric sensors • Documentation & Reporting o Installation reports o Camera and biometric device inventory o Network diagrams o Login accounts and password policies o Maintenance schedules **(Duration- 24 hour)**
	Practical: • **Practical 01 -install, configure, and test CCTV surveillance system.** o Install analog or digital/IP CCTV Cameras o Configure DVR/NVR System o Setup Recording & Storage o Testing & Validation • **Practical 02-install and configure a biometric attendance machine (fingerprint/face ID).** o Install the Biometric Terminal o Configure System Settings o User Enrollment o Data Management **(Duration- 36 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**IT Helpdesk Operations and Technical Support**
Module Code	K72S015M17
Module Type	Core Module
Duration (Hrs.)	60 hour (T-24, P- 36)
Learning Outcomes	After completion of this module the trainee will be able to: 1. Handle IT incidents and service requests efficiently 2. Use helpdesk tools and ticketing systems effectively 3. Troubleshoot and resolve common hardware, software, and network problems 4. Provide professional guidance and support to end-users 5. Maintain accurate records, documentation, and knowledge bases 6. Follow IT security policies and procedures
Learning Content	Theory: • **Introduction to IT Helpdesk** o Role and purpose of IT helpdesk in organizations o Types of helpdesk support: ▪ Level 1 (Basic support) ▪ Level 2 (Intermediate support) ▪ Level 3 (Advanced/Expert support) o Helpdesk responsibilities: ▪ Incident management ▪ Service requests ▪ User guidance and support o Key skills for helpdesk personnel: ▪ Communication skills ▪ Problem-solving skills ▪ Time management ▪ Customer service orientation • **IT Helpdesk Tools and Systems** o Trouble ticket escalating systems o Remote support tools: ▪ Remote desktop access ▪ VPN tools ▪ Screen sharing applications o Knowledge bases and FAQs o Asset management software o Communication channels: ▪ Phone, email, chat, video conferencing • **Incident Management** o Incident logging and categorization o Prioritization of incidents based on severity and impact o Assignment of incidents to appropriate support levels

	○ Escalation procedures ○ Following Service Level Agreements (SLAs) ○ Tracking and monitoring incidents until resolution ○ Updating incident records and closure notes • **Troubleshooting and Problem Resolution** ○ Basic troubleshooting steps: ▪ Identify the problem ▪ Gather information and symptoms ▪ Isolate the cause ▪ Implement solutions ▪ Test and confirm resolution ○ Common hardware issues: ▪ PC/laptop failures ▪ Printer/scanner problems ▪ Peripheral device errors ○ Common software issues: ▪ Operating system errors ▪ Application installation/configuration ▪ Email client issues ○ Networking problems: ▪ Connectivity issues ▪ IP address conflicts ▪ Wi-Fi troubleshooting • **User Support and Guidance** ○ Providing instructions to end-users ○ Assisting users with software applications ○ Educating users on IT policies and security best practices ○ Remote assistance and screen sharing for support ○ Handling difficult or frustrated users professionally • **Documentation and Record-Keeping** ○ Basic call management and recording principles ○ Recording incidents and resolutions in ticketing systems ○ Creating knowledge base on recurring problems and solution history ○ Maintaining asset and configuration records ○ Updating logs for audits and compliance ○ Generating periodic reports for IT management • **IT Policies, Security, and Compliance** ○ Basic understanding of IT policies: ▪ Acceptable use ▪ Password management ▪ Email policies ○ Data security practices:

	▪ Antivirus and malware protection ▪ Secure remote access ▪ Backup and recovery awareness o Compliance and confidentiality: ▪ GDPR, HIPAA, or local IT regulations ▪ Handling sensitive user data • **Performance Monitoring & Continuous Improvement** o Monitoring response and resolution times o Measuring helpdesk performance using KPIs: ▪ Average resolution time ▪ First-call resolution rate ▪ Customer satisfaction rating o Identifying recurring issues and recommending improvements o Feedback collection from end-users for service enhancement **(Duration- 30 hour)**
	Practical: • Practical 01 - Prepare a report with the steps to solve the following problems o PC does not boot (hardware/OS issue) o Printer not printing (driver/network issue) o User cannot access Wi-Fi (network issue) o Application crashes on launch (software issue) • Practical 02 -Document each step in a structured troubleshooting process to solve a problem of a customer/user with the following details: o Problem identification o Gathering information and symptoms o Isolating the root cause o Implementing the solution o Best practices to be followed o Testing and confirming the resolution **(Duration- 30 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Entrepreneurship**
Module Code	K72S015M18
Module Type	Core Module
Duration (Hrs.)	24 hour (T- 12, P- 12)
Learning Outcomes	After completion of this module the trainee will be able to: - Interpret entrepreneurship competencies and success factors - Describe Business launching formalities - Describe Financial administration tasks in a computer maintenance & repairing business - Perform customer awareness for computer maintenance & repairing business - Apply e-waste management aspects
Learning Content	**Theory:** • Introduction to entrepreneurship • Introduction to business plan • Business launching formalities o Business registration and licensing requirements o Business objectives, policies, procedures, strategies and technical guidelines o Legal, regulatory and ethical requirements • Financial tasks in a business o Basic income and expense budgets o Fixed and variable costs o Basic Account charts, weekly basis monthly basis ▪ Cash book, ledger etc. o Knowledge on Current market prices of material and accessories of appropriate quality o Banking procedure • Financial discipline and regular auditing/checking • Pricing structure for various tasks Administration related functions: • Maintaining safe and healthy work environments • Maintaining Customer ’s satisfaction at the reception counter o Pleasant appearance o Courtesy o Mannerism o Empathy • Purchasing procedure • Inventory controlling procedure o Maintaining inventory lists o Re-order procedure

	• HR aspects of team members ○ Keeping personal records ○ Statuary requirements ○ Employee motivation ○ Mentoring /skill upgrading of team members Marketing approach • Goals and objectives • Target market • Branding for the repair center • Positive image built-up • Methods of publicity Green technology aspects • Electronic waste • E-waste disposal procedure **(Duration: 12 Hour)**
	Practical: Assignment-1: prepare a business plan on Entrepreneurship in computer maintenance & repairing business, Business launching formalities, financial tasks in a business, Administration factors, and Marketing approach Assignment-2: brief report on e- waste and disposal **(Duration:12 hour)**
Teaching-Learning activities	• Interactive lecture • Group discussion • Video presentation • Assignments • Demonstrations and practical
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Preparation of Estimation and Quotation**
Module Code	K72S015M19
Module Type	Core Module
Duration (Hrs.)	18 hour (T-12, P- 06)
Learning Outcomes	After completion of this module the trainee will be able to: • Prepare estimate/quotation for new installation / repair work in the computer and peripheral devices / upgrading hardware & software /Preventive maintenance contract or agreement • Finalize the estimate/ quotation with the customer
Learning Content	Theory: • Introduction to costing and estimation principles • Collecting techniques of market data (specifications, prices, availability, validity, delivery, warranty terms) for new installation/repair work/maintenance work • Basic costing and estimation techniques • Material cost, Labour cost, Fixed overhead cost, Traveling cost, Profit margines, Government taxes, Any other additional cost • Introduction to Required validity/timing period • Introduction to Compliances and conformities • Introduction to Supplier registration and tender procedures **(Duration:12 hour)**
	Practical: • Prepare an estimate/ quotation for the customer requirement • Negotiate and finalize the estimate quotation with the customer **(Duration:06 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Communication skills for workplace**
Module Code	K72S015BM01
Module Type	Basic Module
Duration (Hrs.)	12 hour (T-06, P- 06)
Learning Outcomes	After completion of this module the trainee will be able to: - communicate ideas and information at the workplace effectively
Learning Content	Theory: • Listening skills • Telephone etiquettes • Presentation skills • Verbal and non-verbal communication • Safety symbols & procedures • Basic English relevant to workplace and type of work • Organization's rules, regulations and procedures • General norms of polite conduct • Forms of Different manner of address appropriately to customers, superior and subordinates and persons in different positions • Customs and practices of different ethnic and religious groups • Gender issues relevant to the communications • Trends in related technology • Friendliness through a friendly tone, a personal question, or simply a smile, encourage coworkers to engage in open and honest communication • Respect – convey respect for others and their ideas **(Duration:06 hour)**
	Practical: • Role plays and dramas • Assignments 01 - Provide opportunities to trainees to present learned content • Assignments 02 – Prepare list of 250 – 300 technical / vocational terms in English used in relevant industry during the course period • Assignment 03 – Collect manufacturers guidelines/ operational instructions / specifications in relevant field, prepare a file and interpret the information **(Duration:06 hour)**
Teaching-Learning activities	• Assignments • Illustrated talk • Demonstrations • Role play & Drama
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Workshop calculation & Science**
Module Code	K72S015BM02
Module Type	Basic Module
Duration (Hrs.)	36 hour (T-24, P- 12)
Learning Outcomes	After completion of this module the trainee will be able to: • Perform basic arithmetic and trigonometric operations in the workplace when and where required. • Prepare development of surfaces by drawings
Learning Content	Theory • Basic Arithmetic • Area, Volume • Basic trigonometry and angles. • Units and Measurements (Length, Volume, Speed, Time) • Conversion from metric to BS and vice versa • Basic idea of Work, Energy, power & Pressure • Properties of Materials – Basic introduction to Stress, strain, - Definition of Stress, Types of stress- Tensile, compressive, shear, metal expansion and contraction, metal corrosion and prevention • **Heat Treatment Process**– Basic introduction, Definition of heat treatment, Definition of Annealing, Normalizing, Hardening and tempering. Case hardening, • **Units & measurement**: Description, care & use of: - Micrometers- Outside and depth micrometer, Micrometer adjustments, Vernier calipers • **Technical Drawings:** Geometric Construction, preparing development of surface, Concept of true length - Principal methods of development, Development of simple solids like cubes, prisms, cylinders, pyramids, cones. **(Duration:24 hour)**
	Practical • Practice arithmetic and trigonometric operations, • Measuring Practice • Drawing practice **(Duration:12 hour)**
Teaching-Learning activities	• Illustrated talk • Demonstrations • Group and individual practices • Use video clips • Assignments
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Team Work**
Module Code	K72S015BM03
Module Type	Basic Module
Duration (Hrs.)	06 hour (T-00, P- 06)
Learning Outcomes	At the end of this module student should be able to; • Contribute positively to the work in team environment • Work effectively with others in a socially diverse environment • Respect and understand the views of others • Give, receive and act upon feedback • Identify and describe own roles and roles of others
Learning Content	Theory: • Simple concepts of team behavior • Elementary group dynamics • Basic knowledge in social and demographic structures Practical • Identify and establish team purpose, team member's roles and responsibilities. • Coach and motivate team. • Assignment 01- Form a group for workshop cleaning, machine maintenance, waste management and give the feedback on group effort • Assignment 02- Arrange sports activities / social events
Teaching-Learning activities	• Assignments • Lecturers • role plays • case studies • team building activities • coaching and mentoring
Forms of assessment	Formative Assessment + Summative Assessment

Module Title	**Occupational Safety & Health and Environmental Aspects**
Module Code	K72S015BM04
Module Type	Basic Module
Duration (Hrs.)	12 hour (T-06, P- 06)
Learning Outcomes	After completion of this module the trainee will be able to: • Describe importance of safety • Use personal safety gears • Arrange the place to minimize health risk • Give first aid • Recognize and apply environmental aspects in relation with welding process
Learning Content	Theory • Importance of Safety and general Precautions to be observed in the workshop. • Personal safety gears available to protect body; • safety signs - for danger, warning, caution & personal safety message • Safe handling of, Fire extinguishers used for Different types of fire. safe handling and Periodic testing of lifting equipment, Energy Conservation • Types of waste materials • Environmental rules and regulations on disposal of waste (solid & gases) and noise into environment • Standard treatment and disposal methods of waste (solid & gases) and noise • Method of reuse of waste (solid & gases) and scraps **(Duration:06 hour)**
	Practical • Practical related to Safety and Health, • Importance of maintenance and cleanliness of Workshop. • Demonstration on First aid and Fire safety, • Use personal safety gears • Arrange the place of work to minimize the health risks. • Arrange the place of work in order to make it possible to work in the most ergonomic way. • Arrange exhaust system • Give first aid. • Use of fire extinguishers. • Demonstration on safe handling and Periodic testing of lifting equipment • Energy saving Tips

	• Collect details of waste (solid & gases) and scraps in welding process and prepare a brief report with proposal of appropriate disposing/ reusing methods of them **(Duration:06 hour)**
Teaching-Learning activities	• Illustrated talk • Use video clips • Demonstrations • Assignments
Forms of assessment	Formative Assessment + Summative Assessment